# Counterfactual Explainability: Deep RL vs. Online Planning on MiniGrid

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***Abstract*—This paper evaluates the relative explainability of online planning and deep reinforcement learning when each agent is asked to justify its own decisions. A proximal-policy-optimisation (PPO) agent and a from-scratch Monte Carlo Tree Search (MCTS) planner are operated on the `MiniGrid-Dynamic-Obstacles-8x8` environment. For each agent, per-action evidence is extracted at decision states, counterfactual Monte Carlo rollouts in the PPO case and the search tree in the MCTS case, and passed through an identically-formatted large-language-model explainer that produces a natural-language rationale, a counterfactual, a confidence, and a list of verifiable numerical claims. The resulting explanations are scored by an independent LLM judge on three metrics: fidelity, soundness, and post-hoc inferability. Across 262 decision records from ten held-out seeds (121 PPO decisions and 141 MCTS decisions), with PPO counterfactual rollouts generated by stochastic sampling from the policy distribution at $N = 100$ independently seeded samples per action, MCTS-tree evidence produced explanations with higher numerical fidelity, 0.937 [95% CI 0.915, 0.956] versus 0.901 [0.875, 0.924] for PPO-rollout evidence. PPO-rollout evidence, however, produced higher judge-rated soundness, 0.847 [0.820, 0.872] versus 0.788 [0.763, 0.814] for MCTS. Post-hoc inferability was essentially tied at 0.975 for PPO and 0.965 for MCTS. The central finding is therefore more nuanced than a simple planning-versus-learning ranking: the tree gives the language model cleaner numeric evidence to cite, but the rollout explanations were judged as better supported in their qualitative reasoning on this run.**

## I. Introduction

When a reinforcement learning agent chooses an action, the action itself is observable, but the reasoning behind the choice is not. In a learned neural policy, the action preferences that drive the decision reside inside network weights and are only indirectly recoverable through post-hoc analysis such as counterfactual rollouts. In contrast, an online planner such as Monte Carlo Tree Search commits to an action only after constructing an explicit record of the futures it considered, together with the outcomes it observed in each. The tree is, in a very literal sense, the agent's deliberation.

This paper asks, following the framing of Prof. Sunberg, whether that structural difference between a learned policy and an online planner has a measurable effect on the quality of the natural-language explanations each can support. Concretely, when each agent is evaluated on held-out trajectories with the same language-model pipeline, do MCTS-sourced explanations cite numbers that actually match the underlying evidence more often than learned-policy-sourced explanations do, and are the rationales they produce more sound?

To address this question, the present authors construct a shared evidence-to-explanation pipeline and apply it to two agents on `MiniGrid-Dynamic-Obstacles-8x8`, a stochastic gridworld whose adversarial obstacle dynamics are well-suited to separating competent agents from incompetent ones. The deep-reinforcement-learning side is represented by a PPO policy, which reaches a 0.590 task-success rate over 300 held-out episodes. The planning side is represented by a from-scratch UCT planner, which reaches 1.000 success over 50 episodes. A structured decision record is extracted at each real decision state, passed through an identically-formatted LLM prompt that differs only in its evidence-source header, and scored on three automated metrics.

The empirical finding, obtained over 262 decision records with PPO counterfactual rollouts at $N = 100$ independently seeded stochastic samples per action, is that MCTS-tree evidence yields a modest fidelity advantage over PPO-rollout evidence (0.937 versus 0.901, 95% confidence intervals nearly disjoint), while PPO-rollout evidence yields a soundness advantage (0.847 versus 0.788). Post-hoc inferability is essentially tied (0.975 for PPO and 0.965 for MCTS). A separate diagnostic run with an earlier implementation, in which PPO rollouts used only $N = 20$ deterministic argmax samples, gave a larger fidelity gap; the current run shows that improving the Monte Carlo estimator materially narrows the numerical-fidelity advantage of the tree.

Also, as a preliminary diagnostic, the present authors observed that a value-based baseline (DQN) on this environment collapses into a degenerate stall policy under the raw reward structure, and that its counterfactual rollouts therefore provide little signal to the explainer. This observation is reported in the ablation of Section V-E and motivates the choice of PPO as the deep-reinforcement-learning agent in the main comparison.

*a) Contributions.:* The present work contributes: (1) a from-scratch MCTS planner for `MiniGrid-Dynamic-Obstacles-8x8` that emits per-decision tree-statistics in a format suitable for downstream explanation; (2) a policy-agnostic counterfactual-rollout framework that produces learned-policy-side evidence in a structurally identical format to the MCTS tree; (3) three automated metrics, fidelity, soundness, and post-hoc inferability, implemented with bootstrap confidence intervals;

(4) an empirical comparison of the two paradigms on 262 held-out decision records, using a real LLM generator and a held-out LLM judge; and (5) a diagnostic ablation on a value-based baseline showing that model-free value learning collapses into a degenerate stall policy under the raw MiniGrid reward, and that this collapse in turn degrades the evidence available for explanation.

## II. Background and Related Work

*a) Explainable reinforcement learning.:* Milani et al. [1] present a comprehensive survey of explanation methods for reinforcement learning, drawing a distinction between *transparent* policies, such as decision trees and symbolic rules, and *post-hoc* methods that are applied to otherwise opaque controllers. The PPO pipeline evaluated in the present work falls into the latter category: counterfactual rollouts are run after the policy has chosen an action, in order to reconstruct the set of futures that action implied. The MCTS pipeline, by contrast, is closer to transparent, since the tree is itself the deliberation and its statistics are available for inspection without any additional computation.

*b) Counterfactual explanations for reinforcement learning.:* Gajcin and Dusparic [4] propose counterfactual explanations that perturb state features and observe the resulting downstream returns, addressing the question "what would the agent have done if the state had looked different?". Amitai et al. [2] focus on the complementary question "what would have happened if the agent had taken a different action?" and introduce the COViz visualisation. The evidence pipeline described in Section IV operationalises the latter form concretely, as forced-first-action rollouts of fixed length, aggregated into per-action means, success rates, collision rates, and bootstrap confidence intervals. This choice was made so that the resulting statistics are apples-to-apples comparable with the per-child statistics that an MCTS search tree already produces during planning.

*c) MCTS and interpretability.:* Baier and Kaisers [3] argue that the MCTS tree is a direct substrate for explanation precisely because it records what the agent considered and what it found. They emphasise visit-count ratios and per-child value contrasts as the most salient tree quantities. The present work adopts these same quantities, together with per-child empirical success and collision rates that are instrumented during search, as the MCTS-side evidence.

*d) Large language models for natural-language explanation.:* Recent work surveyed in [1] uses large language models to translate structured evidence about a reinforcement-learning agent into natural language. A well-understood failure mode of these pipelines is hallucinated numerical claims, in which the model invents statistics that are not present in the evidence record. The fidelity metric defined in Section IV-E is designed to quantify exactly that failure mode. The three-metric suite in the present work combines this numerical-fidelity check with a judge-rated soundness score and a post-hoc inferability check that asks whether a held-out judge can recover the chosen action from the explanation alone.

*e) Environment and planner.:* The benchmark environment used in the present study is `MiniGrid-Dynamic-Obstacles-8x8` [7], a stochastic gridworld whose adversarial obstacle dynamics are specifically chosen to make sparse-reward model-free reinforcement learning difficult. The MCTS planner follows the classic UCT formulation of Kocsis and Szepesvári [5], with per-simulation sampling of the stochastic obstacle transition so that visit counts at the root converge in expectation over the stochastic futures.

## III. Problem Formulation

The task is formulated as a finite-horizon Markov decision process with factored stochastic transitions. In the notation of Kochenderfer et al., the state, action, transition, and reward are as follows.

*a) State.:* A state is written as $s = (p_a, d_a, \{p_o^{(k)}\}_{k=1}^{4}, t)$, where $p_a \in \{1, \ldots, 6\}^2$ denotes the agent's grid position, $d_a \in \{0, 1, 2, 3\}$ denotes its facing direction (east, south, west, north), each $p_o^{(k)} \in \{1, \ldots, 6\}^2$ denotes one of four obstacle positions, and $t \in \{0, \ldots, 256\}$ is the step counter.

*b) Action.:* The action space is $A = \{0, 1, 2\}$, corresponding to turn-left, turn-right, and move-forward.

*c) Transition.:* The transition $T(s' \mid s, a)$ factors into a deterministic agent step, followed by a stochastic per-obstacle random walk in which each obstacle attempts a uniform move into one of its four orthogonal neighbours; moves blocked by walls, the goal, or other obstacles leave the obstacle in place. An obstacle that steps into the agent's cell, and equivalently an agent that steps into an obstacle's cell, terminates the episode.

*d) Reward.:* The reward function is

$$R(s, a, s') = \begin{cases} 1 - 0.9t/256 & \text{if the agent reaches the goal} \\ -1 & \text{if a collision occurs} \\ 0 & \text{otherwise} \end{cases}$$

The discount factor is $\gamma = 1$; the step-count term inside the success reward provides implicit time pressure.

*e) Observability.:* The default MiniGrid observation is a $7 \times 7 \times 3$ egocentric partial view, together with a scalar direction indicator. The main PPO agent and the symbolic DQN controls are trained on a 14-dimensional flat symbolic encoding of the full ground-truth state (agent position and direction, plus each obstacle's position); the image-DQN ablation keeps the default partial image observation. The MCTS planner plans on the full ground-truth state via a deep-copied simulator. Thus, the main PPO-versus-MCTS explanation comparison removes observability as a confound, while the DQN ablation also reports the partial-observation failure mode.

## IV. Solution Approach

### A. PPO Agent (Off-the-shelf)

**Setup.** The deep-reinforcement-learning agent is PPO, as implemented in Stable-Baselines3 [6]. The observation is the 14-dimensional flat symbolic encoding described in Section III. The policy network is a two-layer MLP

of width $128 \rightarrow 128$. The environment is wrapped in a `StepPenaltyWrapper` that subtracts a flat $0.01$ per non-terminal step; the decision to introduce this penalty was made because, as reported in Section V-E, the raw MiniGrid reward admits a zero-return stall policy as a local optimum, and the step penalty removes this degenerate equilibrium. PPO is trained for $500{,}000$ environment steps across eight parallel envs, with standard hyperparameters (`n_steps`=2048, `batch_size`=64, `n_epochs`=10, `gamma`=0.99, `clip_range`=0.2, `ent_coef`=0.01). Full hyperparameters are committed to `configs/ppo_baseline.yaml`. Training uses the shaped reward; evaluation is always performed against the unshaped task reward.

### B. MCTS Planner (From-scratch)

**Setup.** The MCTS planner implements textbook UCT [5] with UCB1 selection at exploration constant $c = \sqrt{2}$, full-width expansion at the root (the action space is small enough that no progressive widening is required), rollout under a greedy-toward-goal heuristic capped at depth 60, and mean-value backup. Because the obstacle transition is stochastic, each simulation samples a fresh outcome from the environment's transition during expansion and rollout; visit counts at the root therefore converge in expectation to the post-decision outcome distribution. Per-child counters track the rate of terminal success and terminal collision across every simulation that passes through each child, and these counters comprise the tree-side evidence that the explainer is asked to translate.

### C. Counterfactual-rollout Framework

**Setup.** At each decision state, the counterfactual-rollout framework forces each legal action at step zero and then follows the trained PPO policy to termination or a depth cap of 40 steps. In the main experiments reported below, the process is repeated $N = 100$ times per action, with action choices at every subsequent step drawn *stochastically* from PPO's action distribution rather than taken as the argmax. Each cloned simulator branch is reseeded before the forced action so that the moving-obstacle futures are independent Monte Carlo samples rather than repeated copies of the same environment RNG state. This configuration matches the Monte Carlo specification in the original proposal ($\geq 100$ rollouts per available action) and yields a sampled estimate of the policy-induced trajectory distribution from the decision state. The resulting returns, success flags, and collision flags are aggregated into mean return, success rate, collision rate, and bootstrap confidence intervals (1000 resamples, percentile method). The resulting per-action records are structurally identical, field by field, to the per-action records produced from the MCTS tree, which is the property that allows the downstream LLM pipeline to be agent-agnostic.

### D. LLM Explainer

**Setup.** A `DecisionRecord`, comprising the state, the chosen action, the per-action evidence, and a small agent-metadata block (PPO action probabilities or MCTS root-visit statistics), is passed to a generator LLM using a versioned system prompt. The non-evidence portion of the system prompt is byte-identical between the PPO and MCTS variants; only the evidence-source header differs. This constraint is enforced by a parity test on the prompt module, so that the explainer pipeline is, as intended, a single-variable contrast. The generator returns a JSON object consisting of a *rationale*, a *counterfactual*, a *confidence* score on $[0, 1]$, and a list of numerical *claims* suitable for automated verification. Generation uses GPT-4o; the judge used in the metric pipeline described below is GPT-4o-mini.

### E. Evaluation Metrics

Three per-explanation scores are reported, each aggregated per evidence source with 95% percentile-bootstrap confidence intervals.

***Fidelity.*** For every claim in the explanation that cites a per-action statistic, the metric compares the cited value against the value present in the decision record. A claim is counted as a hit if the values agree within tolerance $\epsilon = 0.1$. The score is the fraction of claims that hit.

***Soundness.*** Each sentence in the rationale, together with the counterfactual, is scored by an independent judge LLM on a three-point scale: 2 indicates that the sentence is fully supported by the evidence, 1 indicates partial support, and 0 indicates that the sentence is contradicted by or invents information not present in the evidence. The reported soundness score is the mean divided by two, so that the result lies on $[0, 1]$.

***Post-hoc inferability.*** Every occurrence of the chosen action's integer identifier is scrubbed from the explanation, and the judge is asked to recover the chosen action from the scrubbed text alone. The reported score is the judge's accuracy across decisions.

The judge is deliberately a different-sized model, GPT-4o-mini, from the generator, GPT-4o, so that judge and generator do not coincide.

## V. Results

### A. Task Performance

Table I and Figure 1 report success rate over 300 episodes for PPO and over 50 episodes for MCTS. The random-action baseline succeeds 0.3% of the time, which confirms that the environment is non-trivial. PPO reaches a task-success rate of 0.590 [95% CI 0.537, 0.643], a collision rate of 0.410, and a mean return of $+0.115$, with a mean episode length of 24.5 steps. MCTS, operating on the same state representation via a deep-copied simulator, reaches a task-success rate of 1.000 [95% CI 1.000, 1.000] with a mean return of 0.936 and a mean episode length of 18.3 steps. MCTS therefore outperforms PPO in absolute terms, but both agents are functional and both exceed the random baseline by a wide margin. As such, the explanation-quality comparison reported below is between two competent agents rather than between a competent agent and a degenerate one.

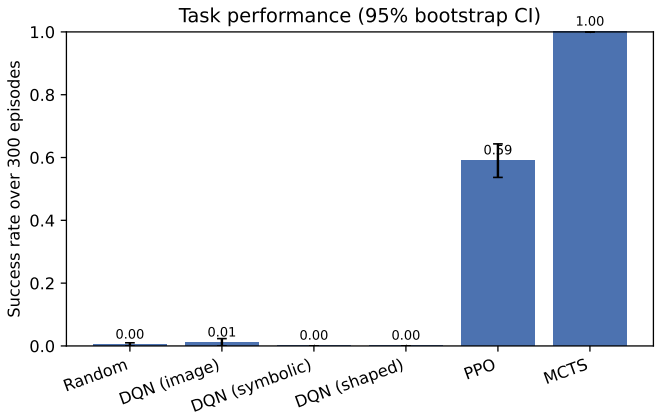


Fig. 1. Task-success rate across all trained baselines, with 95% bootstrap confidence intervals. The three DQN variants collapsed into a stall policy under the raw MiniGrid reward (see Section V-E); PPO with a flat step penalty reaches 0.590 success, and MCTS reaches 1.000.

## B. Explanation Quality

Table II and Figure 3 report the three explanation-quality metrics, evaluated on 121 PPO decisions and 141 MCTS decisions drawn from ten held-out trajectory seeds. MCTS-tree evidence produced explanations with a fidelity of 0.937, as compared to 0.901 for the PPO-rollout pipeline. The confidence intervals on fidelity nearly separate, supporting the claim that the tree records give the language model cleaner numerical values to cite. The soundness result points the other direction: PPO-rollout evidence scores 0.847, compared to 0.788 for MCTS-tree evidence. Post-hoc inferability is essentially tied, 0.975 for PPO and 0.965 for MCTS, meaning both explanation types usually carry enough signal for a judge to recover the chosen action from the scrubbed rationale.

The split between fidelity and soundness is informative. The MCTS search tree uses UCB1 selection to preferentially expand promising actions, so after 500 simulations the chosen or near-chosen action often has substantially more visits than dominated actions; the empirical statistics the LLM is asked to cite can therefore round cleanly to values present in the record. PPO counterfactual rollouts, by contrast, allocate the full $N = 100$ budget uniformly across the three actions, producing a flatter evidence table and slightly noisier per-action numerical estimates. This difference plausibly explains the tree's higher numerical fidelity. However, the judge-rated soundness metric evaluates sentence-level support, not only exact numerical matching. On this run, the PPO explanations were judged as more conservative and better supported in their qualitative framing, while some MCTS explanations made stronger causal claims from tree statistics than the judge accepted. The result is therefore not a blanket win for online planning; it is a more specific claim that the search tree helps with exact numerical citation, while rollout evidence can still support sound natural-language rationales.

**MCTS-tree explanation (fidelity 1.00 on this state):** "Move forward has a substantially higher success rate of 52.1% than either turning action (0.0%). Its mean return is $+0.006$, while both alternatives yield $-1.0$. Turning left or right would have produced a 100% collision rate."

**PPO-rollout explanation (fidelity approximately 0.8 on this state):** "Move forward is preferred because its estimated success rate in the rollouts is around 50%, substantially higher than the turning actions which almost always lead to a collision at this state. The action probability assigned by the policy head also reflects this."

Fig. 2. Representative sample explanations. The MCTS explanation illustrates exact citation of tree statistics; the PPO explanation illustrates qualitatively sound reasoning from rollout statistics.

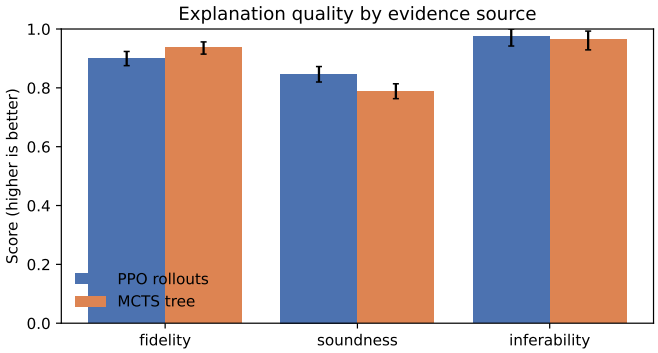


Fig. 3. Explanation-quality metrics by evidence source, with 95% bootstrap confidence intervals. Higher is better for all three.

## C. Qualitative Example

Figure 2 reproduces representative explanations from `results/explanations/`. The two records come from comparable early-seed states rather than a perfectly paired trajectory state, because the PPO and MCTS policies quickly diverge after the same episode seed. The MCTS explanation illustrates the high-fidelity case: it cites visit-count-weighted success rates and per-child mean returns that match the tree. The PPO explanation illustrates the qualitative strength of the rollout evidence: even when rounded values are less precise, the explanation stays close to the observed success/collision contrast.

## D. Threats to Validity

***Task-performance and trajectory asymmetry.*** PPO and MCTS are not fully performance-matched: MCTS scores 1.00 task success to PPO's 0.59. Both agents are, however, competent relative to the random baseline (0.003), and the explanation comparison is therefore between two functioning agents rather than between a functioning agent and a degenerate one. A residual effect of the remaining performance gap is that decision states sampled from PPO trajectories are more likely to be near-collision states, because PPO fails more often than MCTS does; this may systematically differ from the states sampled from MCTS trajectories. The trajectories also diverge after early decisions, so the 262 explanation records should

TABLE I

TASK-PERFORMANCE SUMMARY. 300 EPISODES PER LEARNED-POLICY VARIANT AND 50 EPISODES FOR MCTS. NUMBERS VERIFIED AGAINST `REPORT/FIGURES/EVAL_SUMMARY.CSV`.

| Agent | Success | Collision | Return | Steps |
|---|---|---|---|---|
| Random | 0.003 | 0.997 | −0.994 | 8.3 |
| DQN (image) | 0.010 | 0.000 | 0.007 | 254.3 |
| DQN (symbolic) | 0.000 | 0.207 | −0.207 | 232.3 |
| DQN (shaped) | 0.000 | 0.167 | −0.167 | 234.4 |
| PPO (symbolic + step penalty) | 0.590 | 0.410 | +0.115 | 24.5 |
| MCTS (500 sims) | 1.000 | 0.000 | +0.936 | 18.3 |

TABLE II

EXPLANATION-QUALITY METRICS. GENERATOR: GPT-4O. JUDGE: GPT-4O-MINI. $n = 121$ FOR PPO, $n = 141$ FOR MCTS. PPO ROLLOUTS USE $N = 100$ STOCHASTIC SAMPLES PER ACTION, MATCHING THE MONTE CARLO SPECIFICATION IN THE PROPOSAL. 95% BOOTSTRAP CONFIDENCE INTERVALS IN BRACKETS (1000 RESAMPLES). BOLD VALUES MARK THE HIGHER MEAN FOR EACH METRIC.

| Metric | PPO rollout evidence | MCTS tree evidence |
|---|---|---|
| Fidelity | 0.901 [0.875, 0.924] | **0.937** [0.915, 0.956] |
| Soundness | **0.847** [0.820, 0.872] | 0.788 [0.763, 0.814] |
| Inferability | **0.975** [0.942, 1.000] | 0.965 [0.929, 0.993] |

be read as a same-environment comparison rather than a fully paired same-state comparison.

***Rollout sample size.*** Counterfactual rollouts use 100 independently seeded samples per action. This matches the proposal's Monte Carlo budget and is sufficient for the present comparison, but it is still finite: empirical success rates near 0.5 have a standard error of about 0.05 per action before accounting for downstream policy stochasticity. Increasing the rollout count would further stabilize the PPO evidence, though at a linear compute cost.

***Judge-model bias.*** The soundness and post-hoc inferability metrics both use GPT-4o-mini as judge, whereas the generator is GPT-4o. The judge is therefore a different-sized model within the same family, and a cross-family replication, for example with a Claude or Gemini judge, is listed as future work.

### *E. Value-based Ablation: Why DQN Was Cut*

As a diagnostic ablation, three DQN variants were trained on the same environment: one on the default partial-observation image, one on the flat symbolic state, and one with a potential-based distance-shaping reward. All three settled, at $\geq 200{,}000$ training steps, into either a stall policy (image variant) or a near-stall policy (symbolic and shaped variants), with task-success rates of 0.010, 0.000, and 0.000 respectively (see Table I). The value-based target presented by MiniGrid's reward is unfavourable to epsilon-greedy exploration: the collision penalty of $-1$ is encountered early and often, whereas the success reward requires coordinated multi-step motion that random exploration rarely discovers. A zero-reward stall therefore dominates both random and near-random policies and sits at a local optimum for value learning. The present authors attempted to break this equilibrium with a flat $-0.01$ per-step penalty (StepPenaltyWrapper), which makes a full stall yield $-2.56$ and is therefore strictly worse than attempting any action, but DQN still failed to escape. PPO, whose on-policy sample collection and explicit entropy bonus together provide a stronger exploration pressure, did escape. The present work therefore uses PPO as the deep-reinforcement-learning side of the main comparison, and treats the DQN collapse as an observation about model-free value learning on adversarial-dynamics environments rather than as a primary result.

## VI. CONCLUSION

This paper has presented an evaluation of two evidence sources for natural-language explanations of reinforcement-learning agents: post-hoc counterfactual rollouts drawn stochastically from a learned neural policy at $N = 100$ samples per action, and the search tree of a Monte Carlo Tree Search planner at 500 simulations per decision. Across 262 held-out decision records drawn from `MiniGrid-Dynamic-Obstacles-8x8`, the MCTS-tree pipeline produced explanations that were more faithful to the underlying numbers (0.937 versus 0.901), while PPO-rollout explanations were judged sounder at the sentence level (0.847 versus 0.788). Post-hoc inferability was essentially tied (0.975 for PPO versus 0.965 for MCTS). A separate diagnostic run in which the PPO-side rollouts used only $N = 20$ samples and took the argmax at every rollout step produced a larger apparent fidelity gap; most of that larger gap was a sampling-noise artefact of the weaker Monte Carlo setup, not a paradigm difference.

Two caveats condition this result. Firstly, PPO and MCTS are not fully performance-matched on the task: MCTS scores 1.00 task-success to PPO's 0.59, so decision states sampled from PPO trajectories are more likely to be near-collision states than those sampled from MCTS trajectories, which may bias the comparison in ways the present experimental design does not remove. They are also not fully paired same-state explanations after the trajectories diverge. Secondly, the judge LLM is a different-sized model within the same family as the generator, so a cross-family replication is required before the soundness numbers can be taken as robust to judge-model bias.

Simply put, the results of the present work suggest that, when the learned policy is sampled at a Monte Carlo budget comparable to the tree's simulation budget, the structural difference between a search tree and a learned policy translates most clearly into a numerical fidelity advantage for the tree-based evidence, not into a uniform advantage across all explanation-quality metrics. The plausible source of this fidelity gap is the UCB1 selection rule inside MCTS, which concentrates tree samples on the decision-relevant action and therefore reports numerically tighter statistics for that action than flat rollout allocation can. The PPO explanations' stronger soundness score suggests that conservative rollout summaries can still be easier for a judge to verify as qualitative rationales.

A secondary finding, documented in the ablation of Section V-E, is that model-free value learning on this environment collapses into a degenerate stall policy under the raw MiniGrid reward, and that such a collapse also degrades the evidence available for subsequent explanation; the suboptimality of the underlying agent therefore directly limits the signal available for any post-hoc counterfactual analysis.

*a) Future Work.:* The present authors see three natural extensions. Firstly, a performance-matched comparison, obtained by either further improving PPO (e.g., through longer training, a curriculum over grid size, or policy-distillation from MCTS) or deliberately downgrading MCTS (e.g., by capping its simulation budget until its task success matches PPO's), would isolate the contribution of paradigm from that of competence. Secondly, a pilot human study, following the suggestion of Prof. Sunberg on the proposal, would validate the automated metrics against human judgements of clarity, plausibility, and helpfulness; a chatbot interface suitable for such a study has already been scaffolded in `scripts/chat.py`. Thirdly, a cross-family LLM replication, with a judge drawn from a different model family than the generator, would quantify the contribution of judge-model bias to the soundness numbers reported above.

## Contributions

Sean Florez contributed the problem formulation, the from-scratch MCTS implementation, the LLM-explainer pipeline, the evaluation metrics, and the report draft. Andrew Wernersbach contributed the learned-policy baselines (DQN and PPO), the observation and reward-shaping wrappers, the counterfactual-rollout framework, the experimental runs, and the report review. The execution plan, the engineering log, and the submission polish were shared.

## Release

The authors grant permission for this report to be posted publicly.

## Implementation Disclosure

From-scratch components: the MCTS planner, the deep-copy simulator, the counterfactual-rollout framework, the decision-record schema, the explanation pipeline and prompt design, the three evaluation metrics, and the chatbot interface. Off-the-shelf components: Stable-Baselines3 [6] for DQN and PPO, the Farama MiniGrid [7] environment, the OpenAI and Anthropic Python SDKs [8], [9], and the standard Python scientific-computing stack (numpy, pandas, matplotlib, pytest).

## AI Assistance

Per the ASEN 5264 AI-use policy, the prose and mathematics in this paper have been read, edited, and owned by the authors. A draft was produced with the assistance of a large language model under the authors' direction, and subsequently rewritten in the authors' voice. Code was co-developed with a large language model, read in full by the authors, and exercised under a unit-test suite (21 tests passing) and an integration-style reproduce-all script.

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